Institute of Engineering and Management

E-Commerce Shopping Cart Management System

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PCCCS495 – Term II Project

Introduction

Online shopping has become a major part of modern life. A shopping cart is a core feature of any e-commerce system. This project focuses on building a simple and efficient cart system using Java and OOP

It helps users to:

- Select items
- Modify quantities
- Review products before checkout

Problem Statement

Existing systems often suffer from:

- No validation for invalid inputs
- Poor error handling → system crashes
- Tightly coupled code → difficult to maintain

These issues reduce system reliability and user experience

Target Users

Online Shoppers:
Manage cart items easily before purchase

Gender: Primarily women, aged 18–45.

OOP Concepts Overview

The system demonstrates:

- Abstraction
- Encapsulation
- Polymorphism
- Exception Handling
- Collections (ArrayList)

> **Ensuring clean, reusable and modular code**

@reallygreatsites

OUTPUT VIEW

```
Problems   Javadoc   Declaration   Console ×   Install Java 26 Support
<terminated> OnlineShoppingApp [Java Application] C:\Users\soumy\.p2\pool\plugins\org.eclipse.justj.
Enter your name: Soumyajit

Welcome, SOUMYAJIT
Available Products:
Product ID: S001, Name: Laptop, Price: $899.99, Quantity: 5
Product ID: S002, Name: Smartphone, Price: $599.99, Quantity: 10
Product ID: SS003, Name: Headphones, Price: $49.99, Quantity: 20

1. Add to Cart
2. Remove from Cart
3. View Cart
4. Checkout
5. Exit
1
Enter product id to add:
S001
Enter quantity to add:
5
5 Laptop(s) added to your cart.
Available Products:
Product ID: S001, Name: Laptop, Price: $899.99, Quantity: 5
Product ID: S002, Name: Smartphone, Price: $599.99, Quantity: 10
Product ID: SS003, Name: Headphones, Price: $49.99, Quantity: 20

1. Add to Cart
2. Remove from Cart
3. View Cart
4. Checkout
5. Exit
4
Total Amount to be paid: $4499.95
Thank you for shopping!!!
```

What are the Advantages?

Clean and modular design

Efficient cart management

Prevents duplicate entries

Handles errors gracefully

Easy to extend and modify

Limitations

- No database (data not stored permanently)
- Console-based interface only
- Limited features (no login, no payment system)

Future Enhancements

- Add Graphical User Interface (GUI)
- Integrate database for data persistence
- Implement user authentication system
- Extend to web or mobile application

Conclusion

Successfully developed a functional shopping cart system. This project demonstrates strong use of OOP principles. It provides a reliable and scalable design. The systeman be further enhanced into a complete e-commerce platform.

Thank You